

MicroMAX BEAM CONDITIONING UNIT*

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Abstract

A new Beam Conditioning Unit (BCU) has been developed for the MixroMAX beamline to condition the beam between the KB mirrors and the sample. It includes two XBPMs, a set of slits, a rotating chopper, a fast shutter and a linear attenuator, all on piezo driven stages. MicroMAX has a close collaboration with the BioMAX beamline, to simplify future work the same fastening rail system, with the same distance from rail to beam was chosen. To protect the XBPMs from oxygen but still allow for some heat transfer through convection, the chamber is filled with a low-pressure helium environment.

INTRODUCTION

MicroMAX is a macromolecular crystallography beamline at MAX IV Laboratory designed for time resolved studies of microcrystals in the temperature range between 90 K and room temperature [1, 2].

For MicroMAX a new Beam Conditioning Unit (BCU) has been developed that will manipulate and monitor the beam upstream of the sample position. The BCU will house a commercial X-ray chopper and provide a suitable environment for operation. A compromise between optimized cooling and minimized friction. The beam entering the BCU can either be focused by the Compound Refractive lenses (CRL) or deflected and focused by the Kirkpatrick-Baez (KB) mirrors.

SPECIFICATION

The BCU includes two XBPMs, horizontal and vertical slits, alignment mirror, chopper, fast shutter, scintillator screen, and attenuators. It shall work both with standard beam and with a beam deflected horizontally and vertically by 6 mRad by the KB mirrors. All motion shall be motorized according to the specifications in Table 1.

Table 1: Specifications for Internal Parts

	X	Y	Resolution	Repeatability
XBPM 1 and 2	±5 mm	±5 mm	100 nm	100 nm
Slits	±5 mm	±5 mm	100 nm	100 nm
Alignment mirror	-	-	100 nm	100 nm
Chopper	±5 mm	-	-	-
Shutter and screen	-	-	100 nm	100 nm

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The chopper has a magnetically levitating rotor, so there is no conductive cooling through the motion system, therefore the chamber is filled with low pressure Helium that will transfer some heat without creating too much friction and at the same time protect the XBPMs from oxygen.

DESIGN

For general alignment of the chamber and translation between the deflected and straight beam the BCU will be placed on an external stage covering motion in the vertical and transversal axis plus pitch and yaw motion.

One major design choice was to reuse the rail system used in the European Molecular Biology Laboratory (EMBL) developed BCU at the sister beamline BioMAX [3, 4]. All components except the chopper are assembled on platforms on a rail with the same beam height as in the BioMAX BCU to make it possible to swap parts between the two BCU's if needed. They can be adjusted along the beam, but differently from BioMAX a set of screws are used to give a reference position if a component is taken out for service, see Figs. 1 and 2.

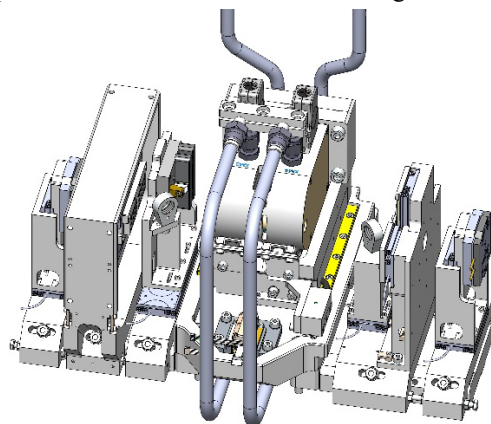


Figure 1: BCU components.

All translations in the BCU are performed by piezo motors and has been selected to fulfil the specifications in Table 1, (see Table 2). The vertical translations for the XBPMs are a high load version to avoid issues experienced at BioMAX. The SLITs are made of four independently controlled tungsten blades originally designed for the NanoMAX beamline. The chopper disc has a step shape to allow for different intervals. To change interval a high load horizontal piezo motor (40 N) is used together with cross roller linear guides. The cooling tubes are long and going around the chopper to allow for more movement and reduce the risk of over bending. The fast shutter and scintillator screen are both on the same horizontal stage, also allowing a passthrough hole in between.

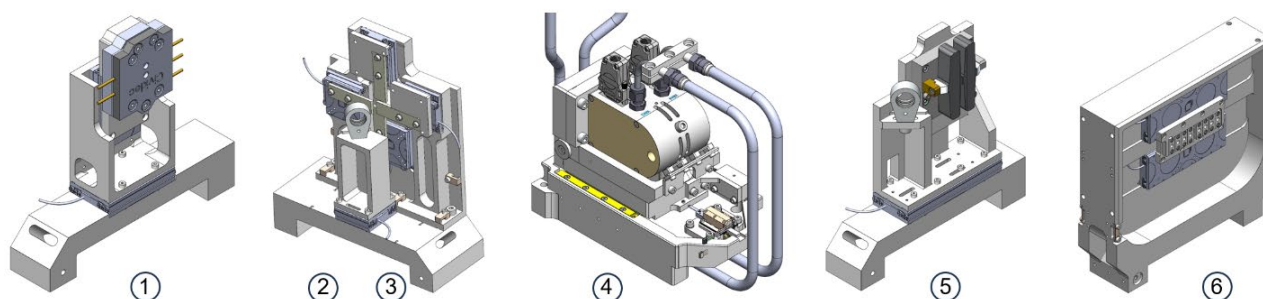


Figure 2: BCU components. 1 and 7: XBPMs, 2: Slits, 3: Alignment mirror, 4: Chopper, 5: Fast shutter and scintillator screen, 6: Attenuators.

Table 2: Translations for Internal Parts as Designed

	X	Y	Resolution	Repeatability
XBPM 1 and 2	± 10 mm	± 15 mm	1 nm	50 nm
Slits	± 10 mm	± 10 mm	1 nm	50 nm
Alignment mirror	-	-	1 nm	50 nm
Chopper	± 8 mm	-	<1 nm	-
Shutter and screen	>50 mm	-	1 nm	50 nm

The attenuators consist of four long piezo stages, each carrying eight small filter holders and a pass-through hole. Allowing a total of 6561 combinations of filters and empty positions.

To transfer heat without causing too much friction in the chopper magnetic levitation bearings a low-pressure helium environment is used. Due to the higher risk of electrical discharges in helium, special precaution was taken for some electrical parts.

The encoder manufacturer we most commonly use and that is used for our chopper stage has made upgrades to their UHV compatible encoders that currently makes them unsuitable for helium atmosphere. Older versions are helium compatible, but in the new one helium enters an internal part causing issues rendering the encoder unusable in helium. To solve this, two spare encoders of the older type were sourced within our facility to use one and have one as spare. To reduce the risk of discharges around the feedthroughs all feedthroughs used are rated >3 A.

Most of the feedthroughs are on the top and back side of the chamber due to equipment restrictions upstream and downstream of the chamber, see Fig. 3, and they are placed so that the equipment inside the BCU doesn't block the feedthroughs, see Fig. 4.

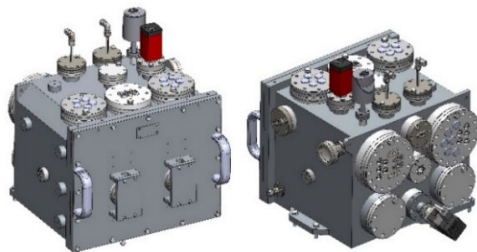


Figure 3: Chamber.

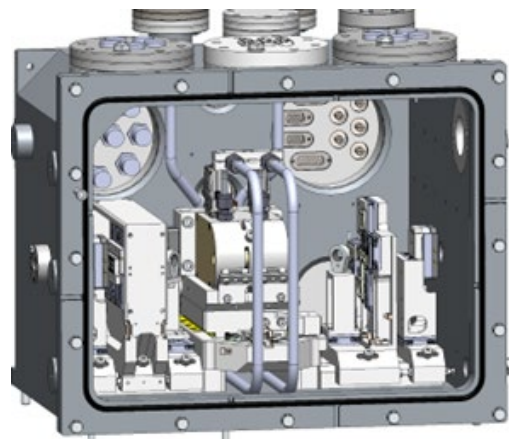


Figure 4: Inside chamber.

PRODUCTION AND INSTALLATION

The chamber was produced externally according to MAX IV drawings and has been installed on a temporary alignment support (see Fig. 5), to be replaced with a motorized four axes stage when the KB mirrors are installed.

The internal mechanical parts were all made in the MAX IV workshop out of mainly aluminium and some parts from stainless steel, assembled by the beamline with silver plated fasteners. Most parts were manufactured with ventilation grooves and ventilated holes. All parts were vacuum cleaned before installation.

Each unit has been assembled individually outside of the chamber and all except the attenuator has been installed in the chamber.

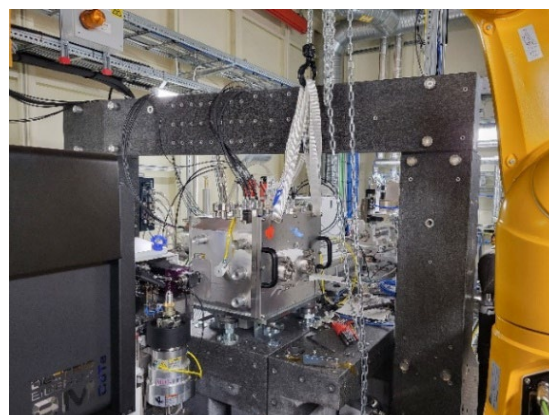


Figure 5: Installation in the chamber.

CONCLUSION

Most parts in the BCU are living up to the expectations and their respective specifications. The chopper stage struggles when moved far in one direction and at one point it seems to have switched in what direction it has issues to the other direction. External loads can be excluded as source of the issue since it runs smoothly without the motor and the motor is strong enough. The issue is still under investigation but likely causes that will be investigated are misalignment of the motor or some residue on the piezo motor shaft. It could also relate to the thin intermediate plate supplied with the motor being too thin and bending, bending has not been detected here. The intermediate plate has been replaced in other beamlines. The attenuators have not yet been installed or tested.

REFERENCES

- [1] MicroMAX, <https://www.maxiv.lu.se/beamlines-accelerators/beamlines/micromax/>
- [2] M. Milas *et al.*, "Experimental station at MicroMAX", presented at MEDSI'25, Lund, Sweden, Sep. 2025, paper TUP27, this conference.
- [3] T. Uhrsby *et al.*, "BioMAX – the first macromolecular crystallography beamline at MAX IV Laboratory", *J. Synchrotron Radiat.*, vol. 27, pp. 1415-1429, Sep. 2020. doi.org/10.1107/S1600577520008723
- [4] M. Cianci *et al.*, "P13, the EMBL macromolecular crystallography beamline at the low-emittance PETRA III ring for high- and low-energy phasing with variable beam focusing", *J. Synchrotron Radiat.*, vol. 24, pp. 323–332, Jan. 2017. doi.org/10.1107/S1600577516016465